Monthly crime trends: a global STL-style decomposition

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What this project estimates

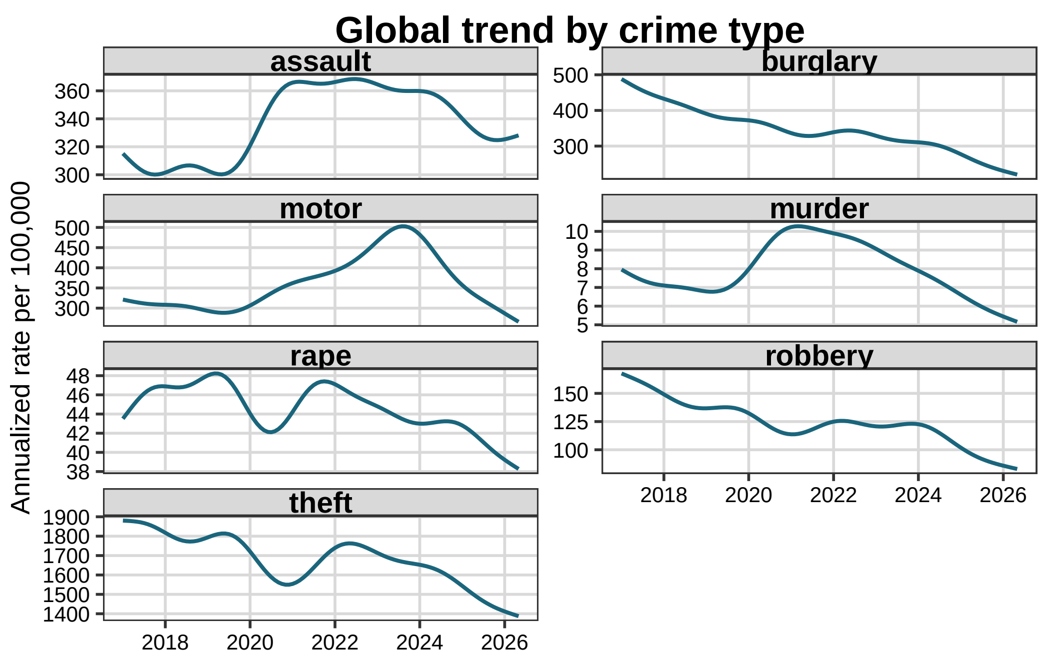
This is a descriptive decomposition of monthly crime counts for every city in the RTCI national sample. Rates are annualized from monthly counts: a monthly count `m` in a population `N` is shown as `12 * m / N * 100,000`. This puts homicide on the familiar annual rate scale rather than below one.

The model estimates a crime-specific smooth long-run trend and a cyclic month-of-year seasonal component for murder, rape, robbery, assault, burglary, theft, and motor-vehicle theft. The global residual is the observed-versus-global departure on the logit scale, centered at zero within crime type. The city-level overdispersion term is retained separately in the complete app data and is centered at zero within city and crime type.

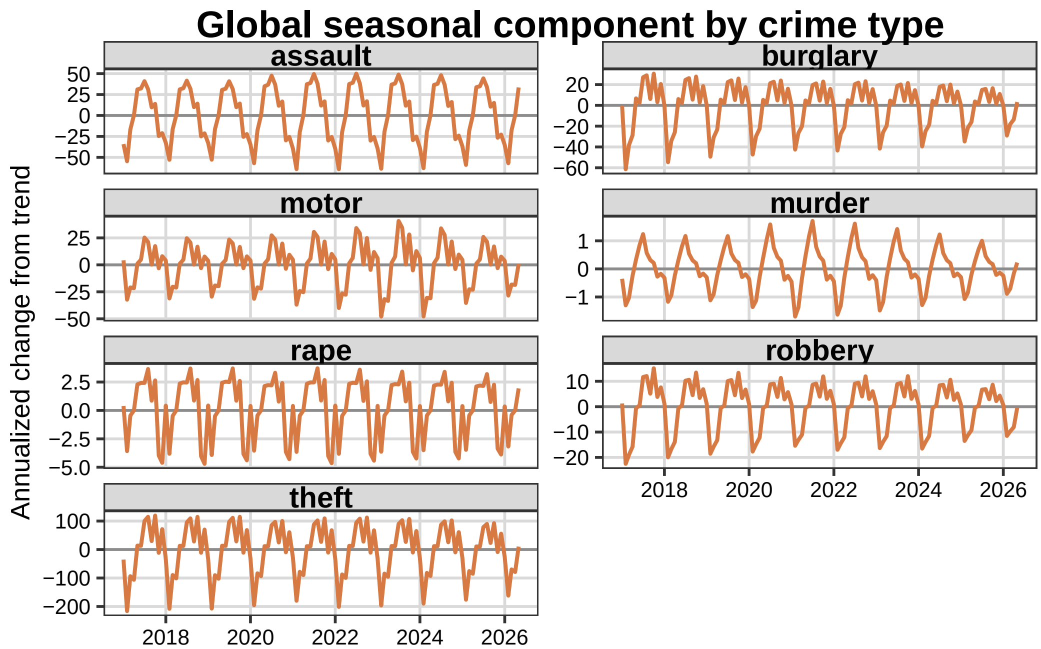
Data coverage

The deliverable contains 458,392 valid city-month-crime observations from 586 cities. There is no population threshold.

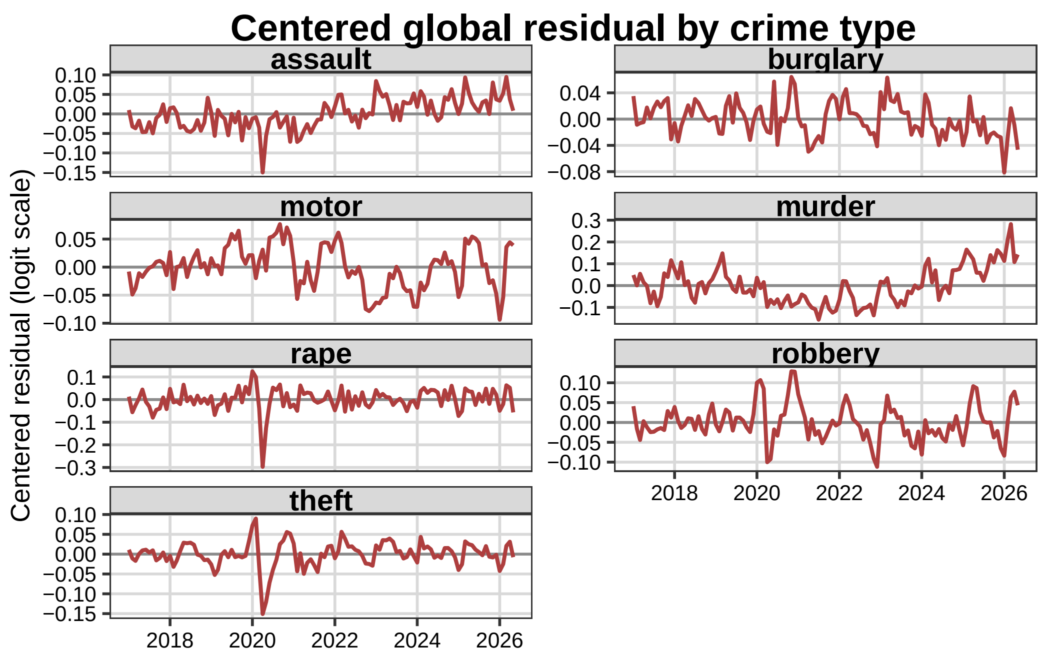
Global trend



Global seasonal component



Global residual



All three figures use one readable panel per crime type. The complete city-by-crime-by-month decomposition and the city-name crosswalk are available in the app data; the app has city and crime selectors plus a map view.